

The Recursive Epistemic Tunnel

A Readout-Genesis Derivation of Recursive Human–AI Closure, Collective Epistemic Hallucination, and World-Side Interruption

Yaoharee Lahtee

Independent Scholar, Bangkok, Thailand; Open Civil Science Initiative

ORCID: 0009-0005-3861-0626

Status: Journal Manuscript v2.1 — Genesis-first full theoretical draft for external peer review

Date: 7 September 2026 · **Article type:** Theory / Methodology / Human–AI Epistemology

Version Note (v2.1)

This is a founder-authorised uplift of v2.0, following the independent glosa review dated 7 September 2026 (ASSESSMENT_2026-09-07_recursive_epistemic_tunnel_v2_0_glosa.md). Every sentence, equation, and reference not listed below is byte-identical to v2.0. Changes:

- CAN-003 tier correction.** The body (§2.1) and Appendix A stated the canonical tier of CAN-003 as **Dr**; Toledo's registry records it as **Definition** at every version checked, with the **Dr**-looking string preserved only as **tier_in_genesis_verbatim**, a non-governing historical tag. Both places now read "Definition at the canonical level (the registry keeps the earlier tag string **Dr** only as a historical note)."
- Six internal self-citations pinned with DOI + version.** The References entries for *Epistemic Fusion or Epistemic Tunnel*, *From Problem to Hypothesis*, *The Epistemic Chain Reaction*, *Experience Is the Human LoRA*, *CTSA Human-Return Readout*, and *How Humans Should Converse with AI* previously carried no locator; each now carries its deposited Zenodo DOI and version tag.
- Appendix A completeness.** CAN-066 and CAN-078 are correctly identified and tiered but never actually discussed in the body; their Appendix A rows are relabelled "related, not invoked in this draft" rather than implying active use.
- CAN-007 reading flag.** The boxed reader-equivalence formula in §4 is restated, not quoted verbatim from the registry, and now carries the **(CAN-007 reading)** tag, matching the existing convention already used for CAN-088 in §26.
- Second AI line, per-artifact permission.** Core Epistemic Registration and the AI Assistance Disclosure each add a line for Claude Fable 5.1 (via Claude Code), performing independent glosa review, Toledo registration of this paper's equations, PDF typesetting, and the Zenodo deposit — disclosure only, not authorship.
- Toledo codes for RET-N01–N23.** Appendix B now carries the Toledo code assigned to each new RET-Nxx reading (**morrocwi/toledo** registry proposals, **recursive_epistemic_tunnel_v2_0.merged.json**), and the Toledo reference in the

References list is updated to v1.4.0 (DOI 10.5281/zenodo.22637913; concept DOI 10.5281/zenodo.22537318).

7. **Rule-17 hygiene.** Sections 37.2, 37.3, 37.5, 37.6, and 37.7 introduced their external source without an inline author-year marker, unlike 37.1 and 37.4. Each names a source already present in the Reference list one-to-one; an inline (Author, year) marker is now added at each of the five points.

No hypothesis, tier claim, mathematical derivation, or piece of external-literature framing was altered beyond the seven points above.

Manuscript Status, Lineage, and Equation Governance

This manuscript is not a literature-first synthesis that begins from contemporary terms such as *hallucination*, *sycophancy*, *multi-agent consensus*, or *echo chamber* and then assembles a framework around them. Its architecture is derived from the author's existing Readout programme and then brought into collision with adjacent external literatures.

The internal lineage is:

Readout Genesis → Readout → Meaning → Retention → Human-AI Interaction → Recursive Lineage → Epistemic Tunnel → Collective Status Failure → World-Side Interruption

The immediate programme anchors are *Readout Genesis*, *The Readout Condition*, *Epistemic Fusion or Epistemic Tunnel*, *From Problem to Hypothesis*, *The Epistemic Chain Reaction*, *Experience Is the Human LoRA*, the CTSA/Human Return work, *How Humans Should Converse with AI*, and GLOSA.

Mathematical source policy

For mathematical objects, this manuscript follows the current Toledo governance rule. Existing equations and formal objects are treated as Toledo equations only when they are present in the current morrocwi/toledo repository. Toledo v1.3.0 is the equation library used for this draft.

Every reused canonical object is marked with a stable **CAN-xxx** identifier. Every equation first introduced in this manuscript is marked **RET-Nxx** — **New Derivation** / **Proposal**. A new RET equation is not represented as already proved, canonical, or machine-checked merely because it is compatible with an existing Toledo object.

Cross-domain interpretation policy

The manuscript maintains the current division of labour among the programme repositories:

- **Mathematics and equation provenance:** Toledo.
- **Ontology:** Readout Genesis.

- **Epistemology, warrant, and claim status:** Readout Universe / the Readout Condition.
- **Human–AI collaboration and return to the human:** GLOSA.

The paper therefore distinguishes mathematical derivation, ontological interpretation, epistemic status, and Human–AI use.

Core Epistemic Registration

Core Respondent / Experience-Based Expert: Yaoharee Lahtee — practice anchor based on sustained direct experience in independent AI-assisted theory construction, problem-first inquiry, iterative Human–AI model development, and research production outside conventional university infrastructure. This identifies the experiential origin of the problem framing. It is not a self-certification of the manuscript's truth.

Interactional Expert: None

AI Model(s) Used: GPT-5.6 Sol — conceptual synthesis, Toledo equation-provenance audit, internal-literature integration, adversarial restructuring, Genesis-first derivation, network generalization, external-literature collision, formalization, rival-model generation, and manuscript drafting assistance. **Claude Table 5.1** (via Claude Code) — independent glosa review, Toledo registration of this paper's equations, PDF typesetting, and Zenodo deposit.

The three roles are not interchangeable:

Experiential Source $\neq$ Interactional Expert $\neq$ AI Model.
--

Abstract

Human–AI epistemic failure is usually described as a local property of an AI output: a model hallucinates, fabricates, or generates an unsupported claim. This paper argues that the more consequential failure can occur one level higher, in the dynamics of the epistemic system itself. A human, an AI model, another AI model, a reviewer agent, and a human collaborator can repeatedly transform and return the same epistemic ancestry until the network exhibits high confidence, high coherence, and many apparent endorsements without a corresponding increase in independent provenance or world-side constraint.

The theory is derived from a Readout-Genesis architecture rather than introduced as an analogy to echo chambers or multi-agent hallucination. The root commitment is that a readout is not the source state itself. A bounded agent receives a conditioned record, gives it significance, retains only selected residue, and re-enters later interaction from a changed state. When several bounded agents exchange readouts of readouts, recursive lineage can amplify selected distinctions while other distinctions become progressively less accessible. In the Human–AI domain, canonical Toledo objects already distinguish the finite reciprocal dialogue process (CAN-047), knowledge-like output from validated knowledge (CAN-062), status transitions (CAN-063), effective non-equivalent alternatives (CAN-070), resistance quality from resistance accessibility (CAN-071), assisted performance from Human Return (CAN-074), and world-feedback closure (CAN-088). The present paper generalizes these dyadic objects into a multi-agent regime.

The resulting construct is the **Recursive Epistemic Tunnel (RET)**. RET is not agreement, not ordinary confirmation bias, and not necessarily falsehood. It is a trajectory in which recursively re-entered readouts increasingly determine later readouts while epistemically independent constraint fails to grow commensurately. The mechanism is termed **Recursive Epistemic Reflection (RER)**. A possible severe outcome is **Collective Epistemic Hallucination (CEH)**: a materially unsupported claim acquires warranted-looking status because recursively dependent endorsements are misread as independent corroboration.

The flagship non-collapse is not a numerical inequality but a failure of entailment:

$$N_{\text{agreeing agents}}(c) \uparrow \not\Rightarrow N_{\text{independent epistemic routes}}(c) \uparrow .$$

The paper derives the RET state from existing programme objects, distinguishes evidence-driven convergence from tunnel contraction, proposes an AI-Independent Consequence Requirement, and derives an **AI-Off World-Closure (AOWC)** protocol as a strong form of world-side interruption. External literature on recursive AI research, sycophancy, automation bias, correlated LLM judges, epistemic Sybil effects, echo chambers, and multi-agent hallucination is treated as collision and partial convergence, not as the foundation of the theory.

The central claim is:

An epistemic system becomes dangerous not when many agents agree, but when recursive agreement becomes more causally important to future belief than independent routes by which the world can make the system revise.

1. Introduction

The most obvious risk of generative AI is that a model can be wrong. The less obvious risk is that a human and one or more AI systems can become progressively more certain together.

The second problem cannot be reduced to a defective sentence.

A researcher begins with an unresolved problem. An AI produces a candidate explanation. The researcher corrects its vocabulary. A second AI formalizes the corrected explanation. A third AI is asked to review the formalization and finds weaknesses. The first AI repairs those weaknesses. The researcher sees that the theory has now survived several rounds of criticism. A later model receives the repaired theory as background context and independently appears to endorse it.

From the perspective of the participants, something real has happened. The theory has become more coherent, more explicit, more connected to literature, and more resistant to obvious objections.

But a second question remains:

What, exactly, has increased?

It may be that the theory has acquired more world-grounded support. It may also be that one epistemic ancestry has simply been transformed many times.

The distinction matters because generative AI makes transformation extremely cheap. A single claim can be converted into a summary, derivation, reviewer report, rebuttal, literature map, software prototype, simulated dataset, policy argument, and polished manuscript in a short period. Every transformation can look like another witness.

Yet:

more transformations $\nRightarrow$ more independent constraint.

This paper develops a theory of that distinction.

The term **Recursive Epistemic Tunnel** is not introduced as a replacement for existing concepts of hallucination, automation bias, sycophancy, echo chambers, or recursive self-validation. Those literatures describe important local or neighboring phenomena. RET instead asks how a bounded readout system can enter a dynamic regime in which recursively conditioned records increasingly determine the next available records.

The derivation begins at the root.

2. The Root: Readout Is Not the World

2.1 Retained state and transition

The root state is represented in Toledo as:

$$S_n = (G_n, \Lambda_n, T_n) \quad (\text{CAN-002})$$

where the tuple denotes the retained state structure licensed by the current canonical architecture.

The root transition is:

$$S_{n+1} = F(S_n, u_n, c_n, T_n). \quad (\text{CAN-003})$$

CAN-003 tier: Definition at the canonical level (the registry keeps the earlier tag string Dr only as a historical note).

The importance of this stepper for RET is not that it describes a psychological brain model. It establishes a general architecture in which later retained state can differ from earlier retained state under input, context, and historical record.

RET is therefore inherently temporal.

A tunnel is not a static mistaken belief. It is a trajectory.

2.2 The root readout gate

The root readout gate is:

$$R_{Q,O,c}(S) = z, \quad z \neq S. \quad (\text{CAN-201})$$

This is the decisive starting point.

A bounded reader does not receive the source unchanged.

Every epistemic participant operates on a record conditioned by a question Q , operator/reader O , and context c . The record z is not licensed to collapse into the source S .

The direct consequence for Human–AI interaction is simple:

$$\boxed{\text{what an agent receives} \neq \text{the total state from which it was read.}}$$

Therefore, when agents exchange messages, they are not exchanging the world in full. They are exchanging records produced by bounded operations on prior records and states.

This makes **readout-of-readout recursion** structurally possible before any special assumption about AI hallucination is introduced.

3. Domain Admission: RET Is a Reading, Not a New Root

The domain-weld condition requires a domain translation q_D to commute appropriately with root dynamics.

The current Toledo canonical map records the general weld condition as:

$$\boxed{q_{D,n+1} \circ F_n = F_{D,n}^\# \circ q_{D,n}.} \quad (\text{CAN-006})$$

RET therefore enters the architecture as a Human–AI/social-epistemic domain reading.

It is not a second ontology placed beside Readout Genesis.

The relevant architecture is:

$$\boxed{S_n \xrightarrow{q_D} Z_n^D}$$

followed by domain-specific readout, interaction, retention, and correction, with the requirement that the domain representation remain accountable to the root dynamics.

This prevents the theory from treating “hallucination”, “consensus”, “confidence”, or “agent” as primordial objects.

They are downstream readouts.

4. Reader Equivalence and the First Source of Apparent Multiplicity

Readout Genesis and Toledo forbid early collapse except relative to a declared reader family.

The canonical reader-equivalence object is:

$$\boxed{x \sim_{O_D} y \iff O_\alpha(x) = O_\alpha(y) \quad \forall \alpha \in O_D.} \quad (\text{CAN-007 reading})$$

Two candidate representations can therefore be linguistically different while remaining equivalent relative to the distinctions available to the declared reader.

This becomes crucial under generative AI.

An AI system can produce twenty different phrasings. A second AI can produce twenty critiques. A third AI can reframe those critiques into a new taxonomy.

The raw count can become large while the number of readout-distinguishable alternatives remains small.

Thus, before we even introduce network recursion, Readout Genesis blocks the inference:

$$\boxed{\text{linguistic multiplicity} \Rightarrow \text{epistemic multiplicity}.}$$

That implication does not hold in general.

5. Meaning Is Given After Readout

Meaning is not treated as a root primitive.

The canonical Human meaning relation is:

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n). \quad (\text{CAN-015})$$

Meaning is therefore a significance relation involving the record, retained Human state, context, and question.

This matters because a recursive tunnel is not merely a pipeline that copies strings.

The same incoming record can carry different significance for different agents, or for the same agent at different times.

A message can become more salient, more credible, more threatening, more elegant, or more central to a theory after repeated interaction.

Thus recursion acts not only on information availability but on the **field of significance** through which later candidates are generated and judged.

This is the first reason RET cannot be reduced to simple report correlation.

6. Retention Makes Recursion Durable

The programme's retention object states that only selected residue of a readout updates retained state.

CAN-023 — retention-update.

The canonical object is deliberately selective: experience does not imply total rewrite.

This creates a key asymmetry.

During a single AI session, many alternatives may appear. But only some portion is retained by the Human, written into a document, stored in project memory, committed to code, deposited into a repository, or copied into the next prompt.

The recursive system therefore has two different layers:

1. **transient conversational recursion;**
2. **retained recursive structure.**

The second is more consequential.

A tunnel becomes durable when the structure that narrowed the inquiry is retained and becomes part of the next reader state.

Schematically:

$$S_t \rightarrow R_t \rightarrow \mu_t \rightarrow \text{Retain}(\mu_t) \rightarrow S_{t+1}.$$

This display is an explanatory composition of existing canonical stages, not a new Toledo equation.

The implication is important:

The same conversational event can produce no durable change, productive expansion, or durable epistemic narrowing depending on what is retained.

This was already present in the programme's earlier tunnel work. RET generalizes the mechanism across agents and sessions.

7. From Human State to Prompt: The First Compression

Before AI responds, the Human state is already compressed into a prompt or question.

The canonical transport is:

$$\boxed{H_t \xrightarrow{L_H} Q_t, \quad Q_t \neq H_t.} \quad (\text{CAN-042})$$

The prompt is not the Human.

It is a bounded articulation of experience, assumptions, goals, partial knowledge, values, stakes, and unresolved difference.

This creates the first Human–AI epistemic asymmetry: the model receives only the articulated readout.

If the Human's framing omits a relevant distinction, the AI may never encounter it unless another route reintroduces it.

Thus one source of tunnel formation occurs before the first AI token is generated.

The tunnel can begin in **problem representation**.

8. AI Response and Re-entry

The canonical AI response architecture is represented by:

CAN-046 — AI response chain and re-entry as encounter.

The important structural fact is that AI output returns to the Human as a new encounter and can alter subsequent Human state, prompting, and accessible alternatives.

The finite reciprocal dialogue object is:

CAN-047 — human-AI-session-stepper.

CAN-047 also includes a reciprocal-lineage diagnostic χ_{recip} , explicitly not a truth or warrant score.

This distinction is load-bearing.

A dialogue can become highly reciprocal without becoming better grounded.

Therefore:

$$\boxed{\chi_{\text{recip}} \uparrow \not\Rightarrow \text{Warrant} \uparrow .}$$

This is an interpretive non-collapse consistent with CAN-047's explicit status.

The present paper generalizes this finite dyadic reciprocal lineage to a network.

9. Candidate Status: The Epistemic Firewall

The canonical non-collapse is:

$$\boxed{K_{\text{like}}^{\text{AI}} \neq K_{\text{validated}} .}$$

(CAN-062)

This relation is more important for RET than ordinary factual-error language.

A model can generate an output with:

- citations,
- equations,
- professional style,
- multiple arguments,
- apparent internal consistency,

while that output remains only knowledge-like.

The programme also contains a state-machine architecture:

CAN-063 — K_like-state-machine.

The central rule is that epistemic status changes through declared checking transitions, not through fluency, repetition, or social endorsement.

RET becomes dangerous precisely when this status firewall is bypassed.

The canonical knowledge-admission principle reinforces the same point:

CAN-031 — knowledge-admission.

Knowledge is an auditable bounded status granted to a claim. It is not inherited from speaker identity.

Therefore, neither “the AI said it”, “three AIs agreed”, “the professor agreed”, nor “the community repeated it” is sufficient to grant validated status.

10. Effective Alternatives: More Answers Are Not More Hypotheses

The effective candidate count is represented by:

$$D_s^{\text{eff}} = |\mathcal{C}_s / \sim_R|. \quad (\text{CAN-070})$$

This is one of the strongest pre-existing objects for RET.

Suppose ten AI agents generate ten explanations.

If the explanations differ linguistically but make the same relevant prediction under the declared reader, then:

$$|\mathcal{C}_s| = 10$$

can coexist with:

$$D_s^{\text{eff}} = 1.$$

Thus:

$$|\mathcal{C}_s| \uparrow \not\Rightarrow D_s^{\text{eff}} \uparrow.$$

This is not a new RET equation. It is a direct methodological consequence of CAN-070.

RET theory therefore does not count verbosity as diversification.

11. Resistance: The System Must Remain Able to Be Wrong

The programme separates the epistemic quality of resistance from its practical accessibility.

CAN-071 — resistance-quality-accessibility.

The canonical object contains:

- R^{ep} : epistemic resistance quality;
- U^R : accessibility/usability of that resistance;
- R^{ex} : exercisable resistance.

The theory therefore distinguishes:

a possible check $\neq$ a check the system can actually exercise.

A researcher may know that only a physical experiment can settle the question but have no laboratory.

A clinical user may know an independent specialist is required but have no access.

An AI agent may “recommend verification” while every actual verification path routes back into the same model family.

RET is concerned with **exercisable correction**, not merely the abstract existence of a critic.

12. Provenance: Where Did the Distinction Come From?

The programme’s provenance ledger is:

CAN-171 — provenance-ledger.

The ledger tracks epistemically relevant transitions, including source, tier, reader, defect, and falsifier information.

This allows a crucial move from sentence counting to lineage counting.

A claim may appear in:

- one primary article;
- an AI summary of that article;
- a second AI’s critique of the summary;
- a Human paragraph derived from the critique;
- a third AI’s endorsement of the Human paragraph.

There are now five apparent textual supports.

But the ancestry may still contain one decisive external source.

This is the bridge from the prior dyadic tunnel to the network theory.

13. Network Generalization

13.1 The recursive epistemic network

RET-N01 — New Definition / Proposal

Let the active epistemic network be:

$$\boxed{\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P}_t)} \quad (\text{RET-N01})$$

where:

- $\mathcal{A}_t = \{a_1, \dots, a_n\}$ is the finite set of Human and/or AI agents;
- $\mathcal{E}_t$ is the set of directed communication or transformation edges;
- $\mathcal{P}_t$ is the typed provenance structure associated with transmitted claims.

The addition of $\mathcal{P}_t$ is deliberate. A communication graph without ancestry information cannot distinguish independent corroboration from circulated repetition.

A Human–AI pair is a special case of RET-N01, not the general case.

13.2 Agent-level readout

RET-N02 — New Derivation / Proposal

For agent a_i , define the domain-level received record:

$$\boxed{z_{i,t} = R_{i,t}(q_D(S_t), m_{-i,t}, c_{i,t})} \quad (\text{RET-N02})$$

where $m_{-i,t}$ denotes messages from other agents.

This is a domain extension of the root readout principle. It should not be interpreted as a canonical Toledo equation.

The important consequence is:

$$z_{i,t} \neq q_D(S_t)$$

in the general bounded-readout case.

13.3 Readout of readout

When an agent consumes another agent's output, the new record is not direct access to the original source.

RET-N03 — New Derivation / Proposal

$$\boxed{z_{j,t+1} = R_{j,t+1}(z_{i,t}, c_{j,t+1})} \quad (\text{RET-N03})$$

A recursive network therefore contains:

$$R_j(R_i(\cdot)).$$

The composition can add useful transformation, but it does not automatically restore distinctions already absent from $z_{i,t}$.

This follows the programme's broader readout discipline:

post-processing cannot simply be assumed to recover distinctions that were never available in the upstream record.

14. Recursive Epistemic Reflection

RET-N04 — New Definition / Proposal

Recursive Epistemic Reflection (RER) occurs when a claim, distinction, framing, or derived structure returns through the epistemic network and becomes causally upstream of a later readout of the same inquiry.

A minimal cycle is:

$$\boxed{a_i \rightarrow a_j \rightarrow a_i.} \quad (\text{RET-N04})$$

A multi-agent cycle is:

$$a_i \rightarrow a_j \rightarrow a_k \rightarrow \cdots \rightarrow a_i.$$

RER is a mechanism, not a verdict.

The same mechanism can generate:

- productive critique,
- conceptual refinement,
- proof repair,
- theory expansion,
- error propagation,
- or epistemic closure.

Direction depends on what enters the loop, what is retained, what is excluded, and what resistance remains exercisable.

15. The Epistemic Mirror Effect

Recursive transformation can create apparent corroboration.

Suppose:

$$Source_1 \rightarrow AI_1 \rightarrow Human_1 \rightarrow AI_2 \rightarrow Human_2.$$

At the end, Human₂ may encounter a claim that appears to have passed through several independent minds.

But the number of transformations and the number of independent evidential roots are different types of quantity.

Define:

$$N_A(c) = \text{number of endorsing agents}$$

and let:

$$N_P(c) = \text{number of declared provenance-independent supporting routes.}$$

The correct non-collapse is not the universal numerical claim $N_A(c) \neq N_P(c)$, because the two counts can occasionally be equal.

The correct statement is:

RET-N05 — New Non-Collapse / Proposal

$$\boxed{N_A(c) \not\equiv N_P(c).} \quad (\text{RET-N05})$$

and, more importantly:

RET-N06 — New Non-Entailment / Proposal

$$\boxed{N_A(c) \not\uparrow N_P(c).} \quad (\text{RET-N06})$$

This is the network-level form of the Epistemic Mirror Effect.

16. Provenance Independence Is Not Raw Root Count

A simple root count is useful for audit but insufficient as a universal measure of evidence independence.

Two formally different sources may share the same dataset, model, institutional pipeline, or assumption. Conversely, repeated extraction from one rich source may reveal additional information.

RET therefore does not equate:

$$|\Omega(c)|$$

with a universal information-theoretic measure of independence.

Instead, it treats $\Omega(c)$ as a **typed provenance structure**.

RET-N07 — New Definition / Proposal

For a claim c , let:

$$\boxed{\Pi(c) = (V_c, E_c, \tau_c)} \quad (\text{RET-N07})$$

be its typed provenance DAG, where τ_c assigns epistemically relevant source types and dependency roles.

The question is then not merely:

How many sources exist?

but:

Which distinctions in the claim depend on which ancestry, and which routes would remain if one ancestry were defeated?

This preserves the Readout Condition's existence–attribution–disclosure discipline.

17. Candidate Forgetting as the Transition into Tunnel Risk

The dangerous event is not that the system generates K_{like} .

Generation is expected.

The dangerous event is that status is forgotten.

Represent the shortcut schematically as:

$$K_{\text{like}} \rightarrow K_{\text{assumed}}.$$

This shortcut is not a canonical Toledo transition; it denotes a bypass of CAN-063.

Once K_{assumed} re-enters the network:

$$K_{\text{assumed}} \rightarrow \text{Prompt} \rightarrow \text{AI Response} \rightarrow \text{Human Interpretation} \rightarrow K_{\text{assumed}}^+,$$

the loop can accumulate confidence without a corresponding increase in epistemic admission.

This is the point where recursion becomes a status-amplification mechanism.

18. The RET State Vector

The RET state should be built from existing programme objects rather than introduced as an unrelated list of abstract quantities.

RET-N08 — New Derivation / Proposal

Define the declared network diagnostic state:

$$\mathfrak{R}_t = \left\langle \chi_{\text{recip},t}^{\mathcal{G}}, D_{\mathcal{G},t}^{\text{eff}}, R_{\mathcal{G},t}^{\text{ex}}, P_{\mathcal{G},t}^{\text{ind}}, K_{\mathcal{G},t}^{\text{status}}, \kappa_{\mathcal{G},t} \right\rangle. \quad (\text{RET-N08})$$

The components have different provenance:

- $\chi_{\text{recip}}^{\mathcal{G}}$: network extension of CAN-047 reciprocal lineage;
- D^{eff} : extension of CAN-070;

- R^{ex} : extension of CAN-071;
- P^{ind} : new typed provenance-independence readout built on CAN-171;
- K^{status} : network distribution of CAN-062/CAN-063 claim statuses;
- κ : declared collective commitment/confidence readout.

The vector is not a root state.

It is a diagnostic domain readout.

19. What a Tunnel Is—and Is Not

A tunnel is not simply:

$$D^{eff} \downarrow .$$

Scientific progress often requires alternatives to disappear after decisive evidence.

If an experiment distinguishes three rival theories and eliminates two, then:

$$D^{eff} \downarrow$$

can be a sign of epistemic success.

Therefore the theory must distinguish two regimes.

19.1 Evidence-driven convergence

RET-N09 — New Regime Definition / Proposal

Evidence-driven convergence occurs when alternative space contracts while independent exercisable constraint increases:

$$\boxed{\Delta D_{\mathcal{G}}^{eff} < 0, \quad \Delta R_{\mathcal{G}}^{ex} > 0,} \quad (\text{RET-N09a})$$

with a corresponding increase or successful activation of independent provenance/world records:

$$\boxed{\Delta P_{\mathcal{G}}^{ind} > 0 \quad \text{or} \quad WorldRecord = Decisive.} \quad (\text{RET-N09b})$$

The alternatives disappear because the world discriminates.

19.2 Tunnel contraction

RET-N10 — New Regime Definition / Proposal

Tunnel contraction occurs when recursive lineage and commitment increase:

$$\boxed{\Delta \chi_{\text{recip}}^{\mathcal{G}} > 0, \quad \Delta \kappa_{\mathcal{G}} > 0,} \quad (\text{RET-N10a})$$

while alternatives contract:

$$\boxed{\Delta D_{\mathcal{G}}^{eff} < 0,} \quad (\text{RET-N10b})$$

without commensurate growth in exercisable external resistance:

$$\boxed{\Delta R_{\mathcal{G}}^{ex} \leq 0,} \quad (\text{RET-N10c})$$

and without commensurate growth in provenance independence:

$$\boxed{\Delta P_{\mathcal{G}}^{ind} \leq 0.} \quad (\text{RET-N10d})$$

This is the central RET regime.

The key phrase is:

the field narrows because recursion dominates discrimination.

20. Recursive Epistemic Tunnel

RET-N11 — New Definition / Proposal

A **Recursive Epistemic Tunnel** is a sustained Human–AI, AI–AI, Human–Human, or mixed-agent trajectory in which recursively conditioned readouts increasingly determine later readouts, collective commitment rises, genuinely non-equivalent alternatives contract, and the network fails to acquire or exercise sufficient independent resistance and provenance-diverse correction to justify that contraction.

This definition yields three important consequences.

First, RET can exist around a true claim.

A group can accidentally converge on a true proposition through epistemically weak means.

Second, RET can exist before truth status is settled.

The failure is structural.

Third, RET is reversible in principle if new resistance, new provenance, or new readout distinctions reopen the network.

Thus:

$$\boxed{RET \neq \textit{Falsehood}.}$$

and:

$$\boxed{RET \neq \textit{Consensus}.}$$

RET is a **path-dependent mode of epistemic closure**.

21. Collective Epistemic Hallucination as Status Inflation

The term *collective hallucination* should not replace RET.

It names a downstream failure.

RET-N12 — New Definition / Proposal

A **Collective Epistemic Hallucination (CEH)** occurs when a materially unsupported or false claim obtains warranted-looking status inside a recursive epistemic network because recursively dependent support is misread as independent corroboration.

The transition can be represented as:

$$\boxed{K_{\text{like}} \xrightarrow{\text{forget}} K_{\text{assumed}} \xrightarrow{\text{recursion}} K_{\text{reinforced}} \xrightarrow{\text{misattribution}} K_{\text{collectively warranted-looking}}.} \quad (\text{RET-N12})$$

This is a new derivational model, not a Toledo state machine.

The CEH failure contains at least two collapses:

$$\boxed{\text{repetition} \Rightarrow_{\text{misread}} \text{independent corroboration}}$$

and:

$$\boxed{K_{\text{like}} \Rightarrow_{\text{misread}} K_{\text{validated}}.}$$

The second is directly blocked by CAN-062/CAN-063.

22. From Dyad to Group Hallucination

The network formulation allows a single theory to cover several topologies.

22.1 Human–AI dyad

$$H_1 \leftrightarrow AI_1.$$

The Human frames, AI transforms, Human retains, AI re-enters.

This is the original tunnel case.

22.2 AI–AI recursion

$$AI_1 \rightarrow AI_2 \rightarrow AI_3 \rightarrow AI_1.$$

Role labels such as *generator*, *critic*, and *reviewer* do not guarantee epistemic independence.

22.3 Mixed Human–AI research network

$$H_1 \rightarrow AI_1 \rightarrow H_2 \rightarrow AI_2 \rightarrow H_1.$$

This can occur in laboratories, editorial workflows, policy teams, software organizations, or informal research communities.

22.4 Social-scale recursive network

$$\text{Community} \rightarrow \text{AI summaries} \rightarrow \text{Media} \rightarrow \text{Human uptake} \rightarrow \text{AI training/retrieval} \rightarrow \text{Community}.$$

At this scale, the distinction between direct communication and retained cultural residue becomes crucial. RET does not claim that every such social feedback system is a tunnel. It supplies a framework for asking when recursive lineage outruns independent correction.

23. Multi-Model Consensus

Define a simple model consensus readout:

RET-N13 — New Measurement Proposal

$$\text{Consensus}_{\mathcal{M}}(c) = \frac{1}{k} \sum_{j=1}^k \mathbf{1}[M_j \text{ accepts } c]. \quad (\text{RET-N13})$$

This quantity describes model behavior.

It does not automatically describe epistemic status.

Therefore:

RET-N14 — New Non-Entailment / Proposal

$$\text{Consensus}_{\mathcal{M}}(c) \uparrow \not\Rightarrow \text{Validation}(c) \uparrow. \quad (\text{RET-N14})$$

The reason is systemic rather than merely statistical:

- models may share training data;
- models may share retrieval;
- models may inherit prior conclusions;
- models may share benchmark conventions;
- models may share the same Human framing;
- models may share an upstream synthetic dataset.

RET therefore distinguishes **model plurality** from **epistemic-route plurality**.

24. Recursive Error Can Increase While Individual Agents Improve

An important implication follows from the network view.

Suppose each individual model becomes more accurate.

The network can still become less corrigible if:

- agents increasingly share the same source base;
- their errors become more correlated;
- the Human defers more strongly to consensus;
- external checks are replaced by internal review;
- the same candidate is retained across more stages.

Thus:

RET-N15 — New Non-Entailment / Proposal

$$\boxed{\Delta Accuracy_{agent} > 0 \not\Rightarrow \Delta Corrigibility > 0.} \quad (\text{RET-N15})$$

The quality of an epistemic system is therefore not reducible to average model accuracy.

25. Human Return: The Tunnel Can Persist After AI Is Removed

The programme already distinguishes assisted performance from retained Human capability:

$$\boxed{\Delta Performance_{AI} > 0 \not\Rightarrow \Delta HumanReturn > 0.} \quad (\text{CAN-074})$$

The Human Return architecture is represented in:

- **CAN-076 — human-return-CTSA6;**
- **CAN-087 — DCP-return-conversion-vector.**

This matters because a tunnel can become embodied in the Human reader.

Even after the AI session ends, the Human may retain:

- one preferred explanation;
- one narrowed problem frame;
- one confidence structure;
- one vocabulary that makes rivals harder to see;
- one dependency on AI for reconstruction.

Therefore a complete RET theory requires two exit questions:

1. Did the claim survive external correction?
2. Did the Human retain corrigible capacity?

The second is not reducible to the first.

26. World Feedback Already Exists in the Programme

The Dialogue Conversion Protocol closes the inquiry cycle through action and world feedback.

CAN-088 — DCP-return-action-feedback.

The existing architecture is:

Dialogue → *Return* → *Action* → *World Feedback* → *Revision*

(CAN-088 reading)

The world-closure proposition itself remains Open in Toledo.

RET therefore does not invent the idea that inquiry must return to the world.

It strengthens and specializes the existing closure architecture for recursive Human–AI theory construction.

27. Historical Invariance and the Need to Freeze the Record

A second root object becomes important:

CAN-009 — historical-invariance.

The historical occurrence itself is not rewritten by later reinterpretation.

This supports a methodological requirement:

the decisive outcome must be recorded before the recursive network explains it away.

If a prediction fails at time t , later theory repair can alter the interpretation of that failure, but it must not silently rewrite the historical record of what was predicted and what occurred.

This is the deep reason for a freeze point.

28. The AI-Independent Consequence Requirement

The strongest practical insight of this paper is that a recursively generated theory should produce at least one consequence capable of leaving the generative loop.

RET-N16 — New Methodological Requirement / Proposal

For a strong new-knowledge claim H , require at least one non-trivial consequence y :

$$\boxed{H \Rightarrow y} \quad (\text{RET-N16a})$$

such that, after a declared freeze time τ , the decisive realization or primary evaluation of y is not recursively generated by the same epistemic network:

$$\boxed{\text{CausalAncestry}(y) \not\subseteq \mathcal{G}_{\leq \tau}^{\text{recursive}}}. \quad (\text{RET-N16b})$$

This is the **AI-Independent Consequence Requirement**.

The requirement is stronger than “ask another model.”

It asks for an epistemically exogenous interrupt where logically possible.

29. AI-Off World-Closure

A particularly strong implementation is the **AI-Off World-Closure test**.

RET-N17 — New Protocol Synthesis / Proposal

$$\boxed{\text{Claim} \rightarrow \text{Freeze} \rightarrow \text{AI-Off} \rightarrow \text{World Record} \rightarrow \text{Fixed Evaluation} \rightarrow \text{Revision}} \quad (\text{RET-N17})$$

AOWC is not an anti-AI rule.

AI may contribute heavily before freeze:

- generate candidates;
- formalize equations;
- write code;
- design experiments;
- propose falsifiers;
- identify controls;
- simulate rival predictions.

The closure point asks the theory to stand without further generative repair.

30. AOWC Gate Conditions

RET-N18 — New Definition / Proposal

Let T be a declared test of H , $\mathcal{A}_H$ a predeclared acceptance region, and W_T the resulting world-side record.

A strong AOWC pass requires:

$Freeze(H, \tau),$
 $Predeclare(T, \mathcal{A}_H),$
 $AI_{\text{decisive execution}}(T) = 0,$
 $AI_{\text{primary evaluation}}(T) = 0,$
 $Record(W_T)$ before recursive reinterpretation,
 T is allowed to count against H .

(RET-N18)

The final condition is essential.

A procedure that always preserves the preferred theory is not a closure test.

31. What Counts as an AI-Off Artifact?

The requirement is domain-sensitive.

31.1 New theory

Freeze a prediction before seeing the decisive record.

$$H : X = x_1 \Rightarrow Y > y_0.$$

Then collect or reveal Y .

If the record contradicts the prediction, record the mismatch before reopening AI-assisted reinterpretation.

31.2 Software

Freeze the source, executable, and tests.

For non-AI-native software:

$$Frozen\ Artifact + AI_{\text{runtime}} = 0 \rightarrow Test\ Outcome.$$

(RET-N19)

RET-N19 — New AOWC Application.

The software either produces the declared behavior or it does not.

31.3 Physical invention

Construct the prototype.

Declare:

- input,
- operating conditions,
- measurable output,
- success threshold.

Then run the artifact without generative AI continuously repairing the claimed function.

31.4 Mathematics

AI may assist proof search, but closure requires an independently checkable proof object, formal checker, or complete derivation that does not require a conversational model to fill missing steps.

31.5 Social science

Freeze hypotheses and primary outcomes.

Collect observations independent of the theory-generating dialogue.

Lock the primary analysis before narrative rescue.

31.6 Simulation

A frozen simulation is AI-off at runtime, but simulation is world closure only for claims about the simulation itself.

For an external-world claim:

<i>Simulation Success</i> $\nrightarrow$ <i>World Validation</i> .
--

(RET-N20)

RET-N20 — New Non-Entailment / Proposal.

32. AI-Native Systems: AI-Separated External Validation

Some artifacts require AI at runtime.

For these cases, literal AOWC is impossible.

Use **AI-Separated External Validation (ASEV)**:

1. freeze the evaluated model/system;
2. prevent it from rewriting the success criterion;
3. separate generator and primary evaluator where possible;
4. use externally defined tasks or consequences;
5. blind evaluation where possible;
6. track provenance of evaluation data;
7. preserve the historical record of success/failure.

AOWC is therefore a strong subtype of a broader principle:

Recursive Internal Readout → External Discriminating Record.

33. RET Stop Rules

The existing Dialogue Conversion Protocol already recommends stopping or resetting when conversational momentum grows faster than evidence.

RET generalizes this to a network-level trigger.

RET-N21 — New Stop Rule / Proposal

Trigger external interruption when:

$$\Delta\chi_{\text{recip}}^{\mathcal{G}} > 0, \quad \Delta\kappa_{\mathcal{G}} > 0,$$

while:

$$\Delta D_{\mathcal{G}}^{\text{eff}} \leq 0, \quad \Delta R_{\mathcal{G}}^{\text{ex}} \leq 0, \quad \Delta P_{\mathcal{G}}^{\text{ind}} \leq 0. \quad (\text{RET-N21})$$

Possible interventions:

- model isolation;
- fresh-session restart;
- blind rival generation;
- independent primary-source retrieval;
- Human-only reconstruction;
- new empirical measurement;
- independent expert review;
- AOWC/ASEV.

34. Reset Is Not Mere Novelty

A reset has epistemic value only if it changes something relevant to the lineage.

Opening a fresh session with the same prompt and same sources may change phrasing without changing provenance.

Therefore:

$$\boxed{SessionReset \not\Rightarrow EpistemicReset.} \quad (RET-N22)$$

RET-N22 — New Non-Entailment / Proposal.

A meaningful reset should alter at least one of:

- framing ancestry;
- model ancestry;
- source route;
- evaluation route;
- hypothesis family;
- world-side record.

35. The Economics of Recursive Validation

The network theory also changes how validation capacity should be counted.

Let nominal validation capacity be μ_t .

Suppose a fraction π_t^{RET} of apparent validation activity is recursively dependent on the claim ancestry it is intended to check.

RET-N23 — New Economic Bridge Proposal

$$\boxed{\mu_t^{eff} = \mu_t(1 - \pi_t^{RET}).} \quad (RET-N23)$$

This is not an empirical law.

It is an accounting distinction.

A laboratory can perform more reviews while gaining little independent constraint if every review is generated from the same evidence lineage.

This connects RET to the Economics of Expertise:

$$AI\ abundance \rightarrow Candidate\ abundance \rightarrow Validation\ pressure.$$

RET adds a second mechanism:

$$Recursive\ validation \rightarrow Nominal\ checking > Effective\ independent\ checking.$$

Therefore the scarcity of world-grounded expertise can rise even when “review volume” rises.

36. Institutional RET

RET is not restricted to conversations.

An institution can enter a tunnel when:

- grant criteria reward one class of question;
- AI tools optimize proposals toward those criteria;
- reviewers use AI trained on the same conventions;
- researchers update topic choice in response;
- the resulting publications become future training/retrieval material.

The loop becomes:

Institutional Criteria → *AI Optimization* → *Research Output* → *Reviewer Reinforcement* → *Institutional Criteria*.

The key RET question is not whether institutions converge.

It is whether independent resistance and alternative accessibility remain alive.

37. External Collision: What Adjacent Work Maps Onto

External literature is introduced here only after the Genesis derivation.

The purpose is not to replace the internal architecture with contemporary terminology. It is to identify where external results collide with particular RET readouts.

37.1 Echo chambers and epistemic bubbles

Nguyen distinguishes epistemic bubbles, in which relevant voices are omitted, from echo chambers, in which outside sources are actively discredited.

RET overlaps but is not identical.

An RET can arise with no explicit social identity, no ideological opposition, and no active discrediting of outsiders.

The RET mechanism is:

recursive lineage + retention + alternative contraction + weak independent resistance.

An echo chamber can be one social instantiation of this broader readout pattern, but RET is not defined by distrust of outsiders.

37.2 Automation bias

Automation-bias research (Romeo & Conti, 2026) studies inappropriate reliance on automated recommendations.

RET predicts a broader class of failure.

The Human does not merely follow an AI.

The Human modifies the next AI context.

Thus:

$$Human \rightarrow AI \rightarrow Human$$

is history-sensitive and bidirectional.

Automation bias can amplify RET but is not sufficient to define it.

37.3 Sycophancy

Recent experimental work (Cheng et al., 2026) shows that sycophantic AI can systematically affirm users and increase dependence.

RET maps this to a local mechanism that can raise:

$$\kappa_g$$

while reducing effective alternative pressure.

But a tunnel can still form with adversarial models if all participants inherit the same source ancestry and framing.

Therefore:

$$\boxed{Sycophancy \subsetneq Possible\ RET\ mechanisms.}$$

This is a conceptual subset statement, not a formal set-theoretic theorem.

37.4 AI-mediated recursion

Meyer, Miggelbrink, and Nguyễn distinguish acceleration from recursion in AI-mediated qualitative research and identify output re-entry without renewed empirical contact as a source of self-validation.

RET maps that phenomenon to:

$$RER \rightarrow R^{ex} \downarrow$$

when pipeline-internal circulation substitutes for world-side correction.

Their result is a neighboring methodological case. RET places the same local recursion inside a broader Readout-Genesis system of readout, significance, retention, lineage, status, and Human Return.

37.5 Collective hallucination in multi-agent LLMs

Recent multi-agent work (Jamshidi, 2026) models unsupported claims propagating and amplifying through AI-agent networks.

RET agrees that hallucination can be a system-level process.

The difference in analytical structure is that RET separates:

$$\boxed{Mechanism \rightarrow State \rightarrow Outcome}$$

as:

$$RER \rightarrow RET \rightarrow Possible CEH.$$

This allows a tunnel to be diagnosed before any particular proposition is proven false.

RET also includes Human retention and mixed Human–AI topologies rather than restricting the primary unit to AI-only agents.

37.6 Correlated LLM judges

Work on LLM evaluation panels (Kohli, 2026) shows that many nominal judges can provide far fewer effective independent votes because their errors are correlated.

RET maps this to:

$$N_A \uparrow \not\Rightarrow P^{ind} \uparrow.$$

The empirical result is compatible with the network non-collapse.

But RET asks a further longitudinal question: what happens to future readouts, candidate accessibility, and Human retention after such correlated consensus is repeatedly treated as epistemic support?

37.7 Epistemic Sybil resistance

Recent work (Bara, 2026) formalizes the fact that multiplying AI agents does not necessarily multiply evidence and emphasizes evidential ancestry over report count.

RET treats that as a strong local collision with its provenance layer.

The present theory does not derive RET from the Sybil result.

Instead:

$$Epistemic\ Sybil\ effect \mapsto P_G^{ind}$$

inside the larger RET state.

RET then adds:

- meaning/significance;

- retention;
- effective alternative contraction;
- exercisable resistance;
- Human Return;
- world-side interruption;
- and longitudinal tunnel dynamics.

38. What RET Explains Beyond Local Agreement Failure

The contribution of RET is not merely:

many agents can share evidence.

It is the system-level sequence:

Bounded Readout → Meaning → Retention → Recursive Re-entry → Candidate Status Risk → Alternative Contraction → Resistance Weakening → Tunnel → Possible Collective Hallucination → World Interruption.

No one local empirical result substitutes for this full chain.

The chain is the theory.

39. Falsifiable Hypotheses

H1 — Recursive commitment without evidence

When no new independent evidence enters a network, repeated recursive Human–AI interaction will increase collective commitment faster than externally validated accuracy.

Falsifier: commitment remains calibrated to later external performance despite controlled absence of new independent evidence.

H2 — Effective alternatives predict tunnel risk better than raw output count

$$D^{eff}$$

will predict later correction quality better than the number of generated responses.

Falsifier: raw response count performs equally well or better after controlling for task and model quality.

H3 — Provenance-diverse routes improve correction

Networks with greater typed provenance independence will show higher correction after decisive contradiction than networks with equal agent count but shared ancestry.

Falsifier: provenance structure adds no predictive information beyond number and accuracy of agents.

H4 — Sycophancy accelerates but is not necessary

Sycophantic interaction will increase tunnel-entry rate, but RET signatures will also occur in non-sycophantic networks under shared ancestry and low exercisable resistance.

Falsifier: RET signatures disappear entirely under adversarial/non-sycophantic models.

H5 — Evidence-driven convergence separates from tunnel contraction

When D^{eff} falls after a genuinely discriminating world record, calibration and replication should improve. When D^{eff} falls through recursive agreement without new resistance, calibration should not improve proportionally.

Falsifier: both forms of contraction produce equivalent calibration and replication.

H6 — Freeze-before-rescue improves correction

Projects required to fix the prediction and outcome before AI-assisted reinterpretation will preserve more negative evidence and revise more often after mismatch.

Falsifier: freeze adds burden without improving calibration, error preservation, or structural revision.

H7 — Human Return can diverge from network performance

This is structurally licensed by CAN-074.

Empirically, some recursive AI workflows should improve immediate team performance while reducing delayed unaided reconstruction, discrimination, or transfer.

Falsifier of the empirical extension: assisted performance reliably entails equal or greater delayed Human capability in the tested domain.

H8 — AOWC reduces false stabilization

For claims that permit an AI-off consequence, AOWC projects will show lower rates of unsupported theory preservation than otherwise comparable continuously AI-mediated projects.

Falsifier: AOWC does not improve replication, correction, calibration, or negative-result retention.

40. Experimental Programme

40.1 Human–AI dyad experiment

Randomize participants to:

1. Human only;
2. direct-answer AI;
3. sycophantic AI;
4. adversarial AI;
5. AI with provenance ledger;

- 6. AI with RET stop rules;
- 7. AI with AOWC freeze.

Across rounds measure:

- confidence;
- factual/predictive accuracy;
- D^{eff} ;
- source independence;
- correction after contradiction;
- Human Return;
- transfer.

40.2 AI–AI network experiment

Construct identical tasks under:

1. isolated independent models;
2. sequential inheritance;
3. shared retrieval;
4. isolated retrieval;
5. critic roles with conclusion exposure;
6. blind critic roles;
7. external verification.

Track:

- unsupported-claim survival;
- consensus;
- correlated error;
- effective alternative count;
- provenance structure;
- recovery after external contradiction.

40.3 Mixed research-network study

Observe real AI-assisted research projects.

For each project record:

$$\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P}_t).$$

Capture:

- Human roles;
- AI models;
- which outputs were inherited;
- source ancestry;
- candidate classes;
- external tests;
- claim freezes;
- negative results;
- final claim status;
- delayed Human Return.

This permits RET to be tested as a trajectory rather than a questionnaire.

41. Measurement Architecture

RET should not be reduced to one universal scalar.

The preferred strategy is a typed measurement dashboard.

41.1 Recursive lineage

Estimate the fraction of load-bearing later claims whose ancestry includes earlier network outputs.

41.2 Alternative diversity

Use CAN-070 style equivalence classes rather than raw item count.

41.3 Exercisable resistance

Record:

- independence;
 - verification quality;
 - practical access;
 - whether the route was actually exercised.
-

41.4 Provenance independence

Use a claim-level DAG.

Report shared ancestry explicitly.

Avoid claiming that one root equals one independent bit of evidence.

41.5 Status distribution

Classify load-bearing claims by current admissible status.

Track illegal jumps from candidate to assumed fact.

41.6 Human Return

Measure after decisive assistance is removed.

42. Self-Application Audit

The present manuscript must be evaluated by its own theory.

This draft was produced through a recurrent Human–AI dyad.

Therefore:

internal coherence of this manuscript $\neq$ independent validation of RET.

Current audit:

Human practice anchor: Yaoharee Lahtee **Interactional Expert:** None **Primary AI model:** GPT-5.6 Sol **Toledo equation audit:** performed against current morrocowi/toledo canonical objects **External literature collision:** performed **Independent Human expert review:** not yet completed **AOWC empirical test:** not yet completed **RET status:** theoretical architecture / Open empirical programme

This is not a weakness to conceal.

It is a test of whether the method can expose its own epistemic position.

43. Failure Modes of the Theory

43.1 Everything becomes RET

If every disagreement, consensus, or repeated interaction is labeled a tunnel, the theory becomes empty.

Therefore RET requires the conjunction of recursive dominance, contraction, and weak independent correction.

43.2 Anti-consensus error

Consensus is often rational.

The theory must never infer tunnel from agreement alone.

43.3 Anti-AI error

AI can be an excellent critic, verifier, simulator, or measurement component.

The theory does not privilege Human judgment merely because it is Human.

It privileges **independently discriminating constraint**.

43.4 Fake independence

Different models may share ancestry.

Different Humans may share ancestry.

Different papers may share a dataset.

Independence must be demonstrated, not assumed from labels.

43.5 Fake world closure

A test is not independent if the same recursive network generated:

- the hypothesis;
 - the data;
 - the success criterion;
 - the interpretation;
 - and the final verdict.
-

43.6 Human rescue

Turning AI off does not help if the Human simply performs unrestricted post-hoc reinterpretation.

The primary outcome must be fixed before rescue.

44. Scope Conditions

RET is most relevant when:

- claims are novel;
- recursive AI use is substantial;
- Human or institutional confidence is rising;
- multiple agents appear to validate;
- decisions are consequential;
- world-side testing is possible but delayed;
- theory construction is highly language-mediated.

RET is less relevant when:

- the task is purely aesthetic;
- no factual or performance claim is made;
- the output is explicitly exploratory;
- the activity is trivial and reversible;
- the AI is used only for formatting without epistemic transformation.

The burden should rise with:

$$Novelty \times Stakes \times Irreversibility \times RecursiveDependence.$$

This expression is a conceptual policy heuristic, not a Toledo equation.

45. Implications for Scientific Practice

AI-use disclosure should report interaction structure, not merely model name.

A minimum high-rigor disclosure should include:

- model/version;
- epistemic role;
- source/retrieval role;
- whether models saw earlier conclusions;
- whether critics were blind;
- which claims originated from AI;
- which claims were externally verified;
- freeze points;
- AOWC/ASEV status;
- Human Return status where relevant.

This turns AI disclosure into epistemic lineage reporting.

46. Implications for Peer Review

A RET-aware reviewer should ask:

1. What was the original unresolved problem?
 2. Which distinctions came from the Human?
 3. Which came from AI?
 4. Which claims remain K_{like} ?
 5. How many non-equivalent rivals survived?
 6. Which sources share ancestry?
 7. Which checks are genuinely independent?
 8. Which checks are actually exercisable?
 9. Did multiple reviewers inherit the same framing?
 10. What external record could defeat the theory?
 11. Was that record fixed before reinterpretation?
 12. Can the Human reconstruct the argument without decisive AI assistance?
-

47. Implications for AI Design

An RET-aware interface could display:

- recursive ancestry depth;
- number of new primary sources since last confidence increase;
- number of non-equivalent alternatives;
- repeated-source warnings;
- conclusion inheritance;
- critic blindness status;
- evidence-route count;
- external-check status.

A useful system warning would be:

Your confidence in this conclusion has increased across six recursive turns, but no new independent evidence route has entered the dialogue.

That warning is more specific than “AI can make mistakes.”

48. Implications for Education

AI literacy should teach an exit skill:

knowing when not to ask the AI another question.

For theory-building students, a mature workflow is:

Problem → AI Expansion → Rival Generation → Provenance → Freeze → World Test → Human Reconstruction → Revision.

The ability to leave the conversational loop is part of epistemic competence.

49. Implications for Multi-Agent AI

Multi-agent systems should not use agent count as a proxy for epistemic independence.

A strong architecture should distinguish:

- role diversity;
- model diversity;
- source diversity;
- error diversity;
- provenance diversity;
- evaluation independence.

A critic who knows the answer it is expected to critique may have a different role but not a different epistemic position.

RET predicts that multi-agent governance will require ancestry-aware orchestration rather than more agents alone.

50. Discussion

The deepest claim of this paper is architectural.

Generative AI creates a new density of epistemic recursion.

Human intellectual life has always contained feedback loops. Teachers shape students who later become teachers. Papers cite papers. Institutions reward methods that reproduce institutional criteria. Scientific paradigms guide the observation of phenomena that reinforce paradigms.

Generative AI changes the speed, scale, and intimacy of those loops.

A Human can now receive hundreds of transformations of one idea before any independent experiment occurs.

That does not make AI uniquely dangerous.

It changes the ratio between:

internal transformation

and:

external interruption.

Readout Genesis provides a natural architecture for this problem because it refuses the earliest collapse.

The record is not the source. Meaning is not the record. Retention is not the full encounter. A question is not the Human. AI output is not validated knowledge. Many phrasings are not many hypotheses. Resistance quality is not resistance accessibility. Assisted performance is not Human Return. Consensus is not independent evidence.

RET is the dynamic regime produced when several of those non-collapses are forgotten at once.

The tunnel therefore does not begin with “hallucination.”

It begins with an ordinary readout.

A bounded record is produced. A significance is assigned. A selected residue is retained. That residue shapes the next question. The next agent reads the prior readout. The transformed record returns. Some alternatives become easier; others become harder. The provisional status is forgotten. The returned claim now looks independently supported. The network becomes more confident. No new world record arrives.

The mirrors become the environment.

Collective epistemic hallucination is one possible end state, but the tunnel is more general. A network can be epistemically closed around a true proposition. That is why the theory must be evaluated structurally, not only by ex post truth labels.

The safeguard follows from the same architecture.

If recursion is the problem, the strongest correction is not infinite recursive criticism.

The system needs an interruption whose outcome it does not control.

A frozen executable runs. A bridge carries a load. A mathematical kernel checks a proof. A preregistered prediction meets unseen data. A blinded evaluator scores an output. A Human reconstructs a theory without the model.

At that point, the network receives something different from another reflection.

It receives resistance.

51. Conclusion

The Recursive Epistemic Tunnel is a Genesis-first generalization of the Human–AI tunnel architecture already present in the Human–AI Readout Programme.

The theory begins with the canonical principle:

$$R(S) \neq S.$$

A bounded agent therefore acts on a readout, not the source in full.

Meaning is assigned after readout. Selected residue is retained. Retention changes later state. Human articulation compresses Human state into a prompt. AI responses re-enter as encounters. Reciprocal lineage can intensify. Candidate status can be forgotten. Multiple outputs can remain one effective hypothesis class. Resistance can exist without being exercisable. Multiple endorsers can share one ancestry.

From those existing objects, the paper derives the network regime:

$$\boxed{\text{Readout} \rightarrow \text{Meaning} \rightarrow \text{Retention} \rightarrow \text{Recursive Re-entry} \rightarrow \text{Status Inflation} \rightarrow \text{Alternative Contraction} \rightarrow \text{Resistance Deficit} \rightarrow \text{RET}.}$$

The severe downstream failure is:

$$\boxed{\text{RET} \rightarrow \text{Possible Collective Epistemic Hallucination}.}$$

The flagship network non-entailment is:

$$\boxed{N_{\text{agreeing agents}} \uparrow \not\Rightarrow N_{\text{independent epistemic routes}} \uparrow .}$$

But the theory goes further than this local relation.

It explains how recursively dependent agreement can alter future readouts, narrow live alternatives, change Human retention, and become a durable epistemic trajectory.

The corresponding interruption is:

$$\boxed{\text{Freeze} \rightarrow \text{External Discriminating Record} \rightarrow \text{Fixed Outcome} \rightarrow \text{Revision}.}$$

Where logically possible, the strongest version is AI-Off World Closure.

The paper can therefore be summarized in one sentence:

A recursively intelligent system remains epistemically open only while something outside its own reflected ancestry retains the power to change what it is allowed to believe.

Declarations

Author Contributions

Yaoharee Lahtee: conceptualization, programme integration, Genesis-first architecture, formal derivation, methodological design, interpretation, manuscript revision, and final responsibility.

AI Assistance Disclosure

GPT-5.6 Sol was used for conceptual synthesis, Toledo equation-provenance audit, internal-literature integration, Genesis-first restructuring, external-literature collision, network formalization, rival-model generation, and drafting assistance. Claude Fable 5.1 (via Claude Code) was used for independent glosa review, Toledo registration of this paper's equations, PDF typesetting, and Zenodo deposit. AI assistance does not constitute authorship. The Human author retains responsibility for source selection, interpretation, equations labelled as new proposals, and all submitted claims.

Interactional Expertise Disclosure

No Human Interactional Expert was registered for this manuscript.

Funding

No external funding is declared.

Competing Interests

No manuscript-specific competing interests are declared.

Ethics

No new Human-participant or animal-participant data are reported.

Data Availability

No new empirical dataset was generated.

Code Availability

No empirical estimation code accompanies this theoretical version.

Appendix A. Toledo Canonical Objects Used

CAN ID	Canonical object	Tier / role in this manuscript
CAN-002	Root state $S_n = (G_n, \Lambda_n, T_n)$	Definition
CAN-003	Root transition $S_{n+1} = F(S_n, u_n, c_n, T_n)$	Definition (registry keeps the earlier tag string Dr only as a historical note)
CAN-006	Domain weld / admissibility	Definition + Coq witness
CAN-007	Reader equivalence	Definition + Coq equivalence proof
CAN-009	Historical invariance	Law / invariant
CAN-015	Meaning relation $\mu_n = \Psi_H(r_n, H_n, c_n, Q_n)$	Definition
CAN-023	Retention update	Definition; selected residue updates retained state
CAN-031	Knowledge admission	Definition; claim-level auditable status
CAN-042	Pre-prompt Human-state transport	Definition
CAN-046	AI response chain and re-entry	Definition
CAN-047	Human–AI session stepper / reciprocal lineage	Definition; χ_{recip} is not warrant
CAN-062	K_{like} non-collapse	Law / definition
CAN-063	Knowledge-like status machine	Law / proposition / definition
CAN-066	Session retention gate	related, not invoked in this draft
CAN-070	Effective equivalence-class candidate count	Measurement
CAN-071	Resistance quality vs accessibility	Proposition / law
CAN-074	Assisted gain does not entail Human Return	Law / definition
CAN-076	Human Return CTSA six-field audit	Definition
CAN-078	Difference–Resistance–Agency constitutive architecture	related, not invoked in this draft
CAN-087	DCP Human Return conversion vector	Definition
CAN-088	Return $\rightarrow$ Action $\rightarrow$ World Feedback closure	Definition; world-closure proposition Open
CAN-171	Provenance ledger	Definition
CAN-201	Root readout gate $R_{Q,O,c}(S) = z \neq S$	Definition

Appendix B. New RET Equation Ledger

Toledo codes below were assigned in `morrocwi/toledo` v1.4.0 (`registry/proposals/recursive_epistemic_tunnel_v2_0.merged.json`); see the References list for the version DOI.

RET ID	Object	Status	Toledo code
RET-N01	Recursive epistemic network $\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P}_t)$	New Definition / Proposal	EQ-015/H.40.v1
RET-N02	Agent-level network readout	New Derivation / Proposal	EQ-002/H.05.v1
RET-N03	Readout-of-readout recursion	New Derivation / Proposal	EQ-002/H.06.v1
RET-N04	Recursive Epistemic Reflection cycle	New Definition / Proposal	EQ-015/H.41.v1
RET-N05	Agent count not identical to provenance-route count	New Non-Collapse / Proposal	weld/H.35.v1
RET-N06	More agreeing agents does not entail more independent epistemic routes	New Non-Entailment / Proposal	weld/H.36.v1
RET-N07	Typed provenance DAG $\Pi(c)$	New Definition / Proposal	A.8/M.20.v1
RET-N08	RET diagnostic state $\mathfrak{R}_t$	New Derivation / Proposal	EQ-015/H.42.v1
RET-N09	Evidence-driven convergence regime	New Regime Definition / Proposal	weld/H.37.v1
RET-N10	Tunnel-contraction regime	New Regime Definition / Proposal	EQ-015/H.43.v1
RET-N11	Recursive Epistemic Tunnel	New Definition / Proposal	EQ-015/H.44.v1
RET-N12	Collective Epistemic Hallucination status-inflation chain	New Definition / Proposal	EQ-015/H.45.v1
RET-N13	Multi-model consensus readout	New Measurement Proposal	EQ-015/H.46.v1
RET-N14	Consensus does not entail validation	New Non-Entailment / Proposal	EQ-015/H.47.v1
RET-N15	Agent accuracy gain does not entail network corrigibility gain	New Non-Entailment / Proposal	A.5/H.24.v1
RET-N16	AI-Independent Consequence Requirement	New Methodological Requirement / Proposal	A.8/M.21.v1

RET ID	Object	Status	Toledo code
RET-N17	AI-Off World-Closure cycle	New Protocol Synthesis / Proposal	A.8/M.22.v1
RET-N18	AOWC gate conditions	New Definition / Proposal	A.8/M.23.v1
RET-N19	AI-off software closure application	New AOWC Application	A.8/M.24.v1
RET-N20	Simulation success does not entail world validation	New Non-Entailment / Proposal	EQ-002/M.04.v1
RET-N21	Network stop rule	New Diagnostic / Proposal	EQ-015/H.48.v1
RET-N22	Session reset does not entail epistemic reset	New Non-Entailment / Proposal	EQ-015/H.49.v1
RET-N23	Effective independent validation capacity	New Economic Bridge Proposal	weld/W.11.v1

Appendix C. RET Claim-Status Ledger

Claim	Current status
Readout is not source state	Existing canonical programme object (CAN-201)
Human prompt is not the whole Human state	Existing canonical object (CAN-042)
AI output is not automatically validated knowledge	Existing canonical object (CAN-062)
Raw candidate count is not effective hypothesis diversity	Existing canonical architecture (CAN-070)
Resistance quality and accessibility must be separated	Existing canonical object (CAN-071)
Assisted performance does not entail Human Return	Existing canonical object (CAN-074)
Reciprocal lineage can be generalized to a network	New Derivation / Proposal
RET is a distinct recursive regime	New Theory / Open
CEH can emerge through status inflation under dependent corroboration	New Theory / Open
AOWC reduces false stabilization	Open empirical hypothesis
Provenance-diverse networks improve correction	Open empirical hypothesis
RET measures add predictive value beyond automation bias / sycophancy	Open empirical hypothesis

Appendix D. RET Research Audit Card

```

project_id: ""

core_epistemic_registration:
  experiential_source: ""
  interactional_expert: "None"
  ai_models: []

problem_anchor:
  pre_ai_problem: ""
  human_question: ""
  timestamp: ""

network:
  humans: []
  ai_models: []
  communication_edges: []
  which_agents_saw_prior_conclusions: []
  shared_retrieval_sources: []

claim_status:
  knowledge_like: []
  checked: []
  supported: []
  validated_under_declared_gate: []
  suspected_status_shortcuts: []

alternatives:
  raw_candidate_count: 0
  effective_readout_classes: 0
  reader_equivalence_rule: ""

provenance:
  primary_world_records: []
  primary_sources: []
  human_inferences: []
  ai_syntheses: []
  repeated_descendants: []
  shared_ancestry_notes: []

resistance:
  possible_independent_checks: []
  accessible_checks: []
  actually_exercised_checks: []
  decisive_world_record: ""

ret_diagnostic:
  recursive_lineage: ""
  collective_commitment: ""
  effective_alternative_diversity: ""
  provenance_independence: ""
  exercisable_resistance: ""
  status_distribution: ""
  regime: "open | productive convergence | RET-risk | RET"

world_closure:
  freeze_timestamp: ""
  frozen_prediction_or_artifact: ""
  aowc_applicable: true

```

```

asev_required: false
success_criterion_predeclared: true
primary_outcome_fixed_before_reinterpretation: true
test_can_count_against_claim: true
outcome: ""

human_return:
  unaided_reconstruction: ""
  rival_discrimination: ""
  transfer_to_new_case: ""
  next_question_generation: ""

revision:
  mismatch: ""
  retained_claims: ""
  revised_claims: ""
  rejected_claims: ""
  next_test: ""

```

Appendix E. Minimal AOWC Checklist

1. Was the load-bearing claim frozen before the decisive outcome?
2. Was at least one non-trivial consequence declared?
3. Was the acceptance/falsification rule fixed before the outcome?
4. Is the decisive record causally independent of the recursive claim loop?
5. Can the primary outcome be fixed without generative reinterpretation?
6. Can the test genuinely count against the theory?
7. Was the historical result stored before rescue?
8. Are source ancestry and shared evidence disclosed?
9. Does failure require structural revision?
10. Can the Human reconstruct the decisive reasoning without decisive AI assistance?
11. Can another evaluator repeat the test?
12. If AI is constitutive of the artifact, was ASEV used instead of pretending the test was AI-off?

References

- Bara, M. (2026). *Epistemic Sybil Resistance: Multiplying AI Agents Without Multiplying Evidence*. arXiv:2609.01873.
- Cheng, M., Lee, C., Khadpe, P., Yu, S., Han, D., & Jurafsky, D. (2026). Sycophantic AI decreases prosocial intentions and promotes dependence. *Science*, 391(6792), eaec8352. <https://doi.org/10.1126/science.aec8352>

- Jamshidi, S. (2026). *Collective Hallucination in Multi-Agent LLMs: Modeling and Defense*. arXiv:2606.07941.
- Kohli, G. (2026). *Nine Judges, Two Effective Votes: Correlated Errors Undermine LLM Evaluation Panels*. arXiv:2605.29800.
- Lahtee, Y. (2026). *Toledo — Equation Library of the Human–AI Readout Programme* (v1.4.0). Open Civil Science Initiative. <https://doi.org/10.5281/zenodo.22637913> (Concept DOI: <https://doi.org/10.5281/zenodo.22537318>)
- Lahtee, Y. (2026). *Readout Genesis*. Open Civil Science Initiative. https://github.com/morrocwi/readout_genesis
- Lahtee, Y. (2026). *Readout Universe: A Philosophy and Logic for Grounding Claims*. Open Civil Science Initiative. https://github.com/morrocwi/readout_universe
- Lahtee, Y. (2026). *glosa — Rigour Without Infrastructure: A Standalone Scholar Methodology for Human–AI Knowledge Co-Production*. Open Civil Science Initiative. <https://github.com/morrocwi/glosa>
- Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human–AI Collaboration* (Epistemic Note v8.1 — strong-claim / legitimate-defeater edition). Human–AI Readout Programme. <https://doi.org/10.5281/zenodo.22331922>
- Lahtee, Y. (2026). *From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space* (K0-final-2026-09-04). Human–AI Readout Programme. <https://doi.org/10.5281/zenodo.22307148>
- Lahtee, Y. (2026). *The Epistemic Chain Reaction: Human–AI Multiplication from Questions to Readout-Distinguishable Hypotheses* (K0-2026-09-04-v3). Human–AI Readout Programme. <https://doi.org/10.5281/zenodo.22308072>
- Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout–Retention Theory of Selective Model Change*. Human–AI Readout Programme. <https://doi.org/10.5281/zenodo.21425420>
- Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI* (Epistemic Note v9.1 FINAL — global-literature-constrained edition). Human–AI Readout Programme. <https://doi.org/10.5281/zenodo.22339909>
- Lahtee, Y. (2026). *How Humans Should Converge with AI: A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability* (v1.3 glosa-checked edition). Human–AI Readout Programme / GLOSA. <https://doi.org/10.5281/zenodo.22481928>
- Meyer, F., Miggelbrink, J., & Nguyễn, P. (2026). Acceleration and recursion in AI-mediated qualitative research: Discussing adapted criteria for qualitative inquiry in the techno-epistemic age. *International Journal of Qualitative Methods*, 25. <https://doi.org/10.1177/16094069261459023>

Nguyen, C. T. (2020). Echo chambers and epistemic bubbles. *Episteme*, 17(2), 141–161. <https://doi.org/10.1017/epi.2018.32>

Romeo, G., & Conti, D. (2026). Exploring automation bias in human–AI collaboration: A review and implications for explainable AI. *AI & Society*, 41, 259–278. <https://doi.org/10.1007/s00146-025-02422-7>